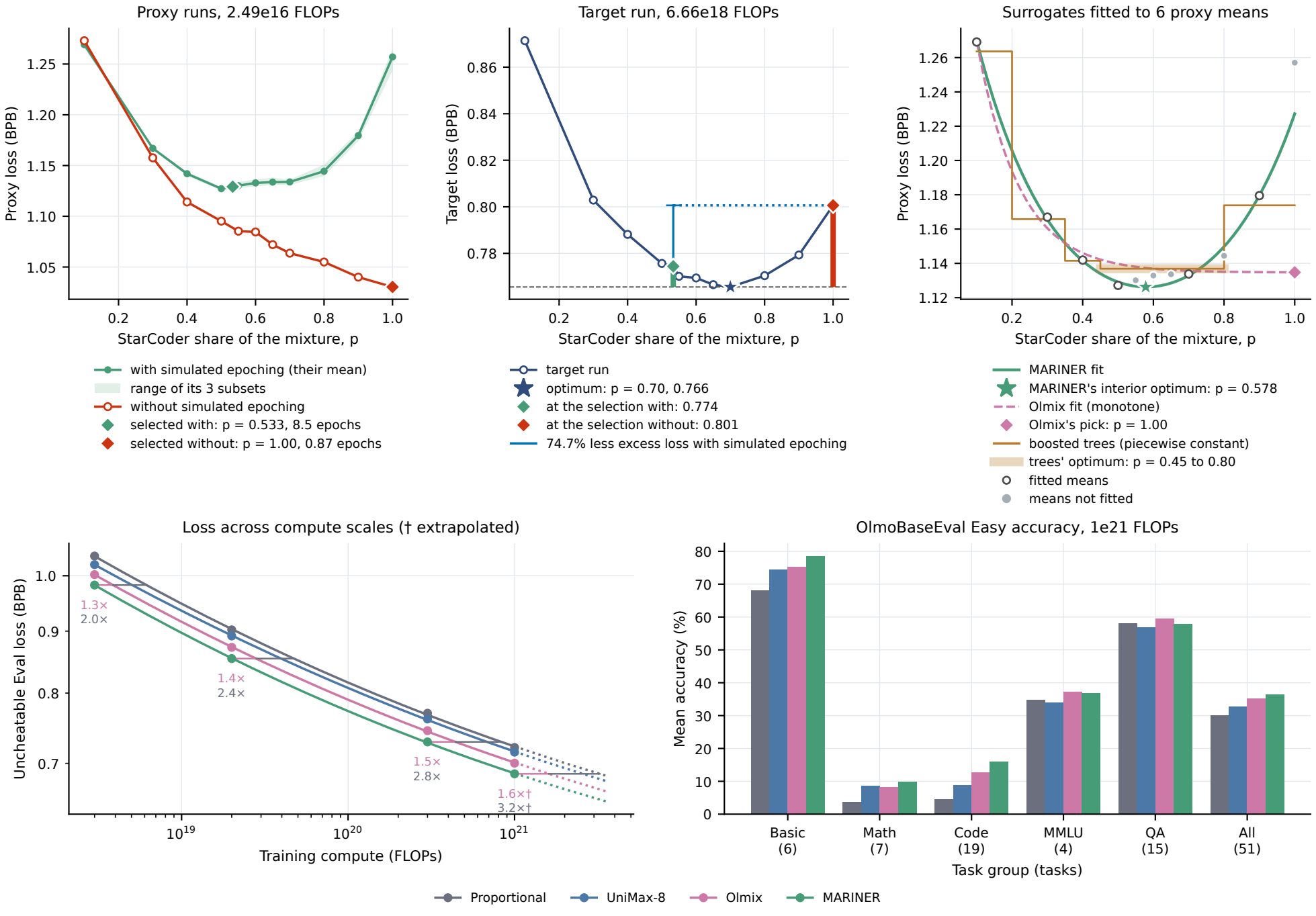
Proxy runs, 2.49e16 FLOPs
Proxy loss (BPB)
StarCoder share of the mixture, p
with simulated epoching (their mean)
range of its 3 subsets
without simulated epoching
selected with: p = 0.533, 8.5 epochs
selected without: p = 1.00, 0.87 epochs
Target run, 6.66e18 FLOPs
Target loss (BPB)
StarCoder share of the mixture, p
target run
optimum: p = 0.70, 0.766
at the selection with: 0.774
at the selection without: 0.801
74.7% less excess loss with simulated epoching
Surrogates fitted to 6 proxy means
Proxy loss (BPB)
StarCoder share of the mixture, p
MARINER fit
MARINER's interior optimum: p = 0.578
Olmix fit (monotone)
Olmix's pick: p = 1.00
boosted trees (piecewise constant)
trees' optimum: p = 0.45 to 0.80
fitted means
means not fitted
Loss across compute scales († extrapolated)
Uncheatable Eval loss (BPB)
1.3×
2.0×
1.4×
2.4×
1.5×
2.8×
1.6×†
3.2×†
Training compute (FLOPs)
OlmoBaseEval Easy accuracy, 1e21 FLOPs
Mean accuracy (%)
Basic (6)
Math (7)
Code (19)
MMLU (4)
QA (15)
All (51)
Task group (tasks)
Proportional
UniMax-8
Olmix
MARINER